Fikile Mkhize 0210210785080

0781087624

14 July 2026

REPRESENT ANALYSE AND CALCULATE SHAPE AND MOTION IN 2 AND 3 DIMENSIONAL SPACE IN DIFFERENT CONTEXTS

US9016

Summative assessment

Question 1

1. A: Protractor

-B: Thermometer

- C: Weighing scale / Kitchen scale

-D: Steel ruler

- E: Tape measure / Measuring tape

2.A. The pointer is on 80° from the left,110° from the right

- B: Using °C scale: The line is between 20°C and 30°C, closer to 24°C. Reading = 24°C

Using °F scale: Between 70°F and 80°F = 74°F

-C:The needle points to 500g

- D: On the steel ruler it reads approximately 3 mm

-E: The tape measure shows the object is 10 inches

long

3.- A:Angle → radian (rad).Commonly used: degree (°)

-B:Temperature → Kelvin(K).Commonly used: °C

- C: Mass → kilogram (kg) and gram (g)

-D:Length → metre (m) and mm is also used

- E:Length → metre (m)

Qestion 2

a) Use the water displacement method / Archimedes principle

1. Fill a measuring cylinder/container with a known volume of water.

2. Carefully submerge the vase fully in the water.

Fikile Mkhize 0210210785080

3. Measure the new water level.

4. Volume of vase = Final water volume - Initial water volume.

b)Formula: Volume = Length × Width × Height

i. Square box: 200mm × 200mm × 200mm

V = 200× 200 × 200 = 8 000 000 $mm^3$

ii. Rectangular box: 80mm × 80mm × 800mm

V = 80×80 × 800 = 5 120 000 $mm^3$

The square box carries more water. 8 000 000 $mm^3$ > 5 120 000$mm^3$

Question 3

Formula: Time = Distance/Speed

a)Time = 60 miles/40 m/h= 1.5 hours

= 1 hour 30 minutes

b)Bill's time =80 miles/40 m/h= 2 hours

Difference = 2 - 1.5 = 0.5 hours

= 30 minutes longer

Question 4

Ship goes 300km West, then 100km South. Forms a right triangle.

Distance from start = hypotenuse

c = \sqrt($a^2$+ $b^2$)= \sqrt($300^2$+ $100^2$)

c = \sqrt{90000 + 10000} = \sqrt{100000}$

c = 316.23 km

316.23 km from starting position *

Question 5

Rectangle: 10m by 16m

Path A: Walk along 2 sides = 10 + 16 = 26 m

Path B: Diagonal = \sqrt($10^2$+ $16^2$) = \sqrt(100 + 256)= \sqrt(356)= 18.87 m

Difference: 26 - 18.87 = 7.13 m

Ron walks 7.13 m further when his dad is around.

Fikile Mkhize 0210210785080

Question 6

Diameter = 50m, so radius r = 25m.Height h = 30m

I. Volume of cone

$V = (1/3)\pi r^2 h$

$= (1/3) \times 3.142 \times 25^2 \times 30$

$V = \frac{ \times 3.142 \times 625 \times 30 = 76\,576.32\ m^3$

II. Surface area of the wall = Curved Surface Area

Slant height $l = \sqrt{r^2 + h^2} = \sqrt{25^2 + 30^2} = \sqrt{625 + 900} = \sqrt{1525} = 39.05$ m

$CSA = \pi r l = 3.142 \times 25 \times 39.05 = 3\,067.32\ m^2$